

ECE 750T Assignment 1

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Question 1

Q1.1

On the training set, ranking by absolute Pearson correlation and breaking ties by column index gives the following signed correlations.

Feature correlated with Y	Correlation
marital-status_Married-civ-spouse	0.4447
relationship_Husband	0.4010
education_num	0.3352
marital-status_Never-married	-0.3184
age_u30	-0.2381
hours-per-week	0.2297
relationship_Own-child	-0.2285
capital-gain	0.2233
occupation_Exec-managerial	0.2149
relationship_Not-in-family	-0.1885

Feature correlated with A	Correlation
relationship_Husband	0.5801
marital-status_Married-civ-spouse	0.4318
relationship_Unmarried	-0.3213
relationship_Wife	-0.3193
occupation_Adm-clerical	-0.2631
hours-per-week	0.2293
marital-status_Divorced	-0.2286
occupation_Craft-repair	0.2231
marital-status_Widowed	-0.1885
marital-status_Never-married	-0.1714

Q1.2

The supplied classifier, trained with $\alpha = 0$, has test accuracy 0.8557, reweighted accuracy 0.8738, and $\Delta_{DP} = 0.1707$. The positive-prediction rate is 0.0880 for $A = 0$ (Female) and 0.2587 for $A = 1$ (Male), so Male has the higher average prediction.

Q1.3

The three strongest test-set correlations with $\hat{Y}$, again ranked by absolute correlation, are:

Subset	Feature	Correlation
All	marital-status_Married-civ-spouse	0.4686
All	education_num	0.4336
All	relationship_Husband	0.4177
$A = 0$	relationship_Wife	0.5620
$A = 0$	marital-status_Married-civ-spouse	0.5273
$A = 0$	capital-gain	0.2987
$A = 1$	education_num	0.4913
$A = 1$	relationship_Husband	0.4151
$A = 1$	marital-status_Married-civ-spouse	0.4105

Q1.4

(a) For a batch containing both groups, let $r_i \in (0, 1)$ be the sigmoid score and $B_a = \{i : A_i = a\}$. The penalty is

$$L_{\text{DP}} = \left(\frac{1}{|B_0|} \sum_{i \in B_0} r_i - \frac{1}{|B_1|} \sum_{i \in B_1} r_i \right)^2.$$

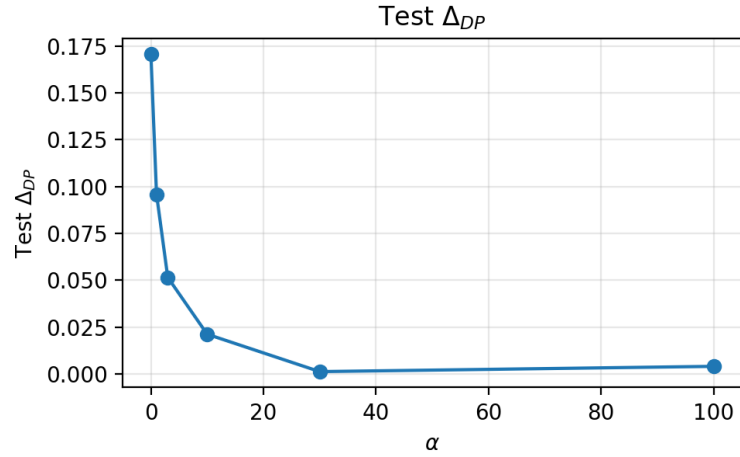
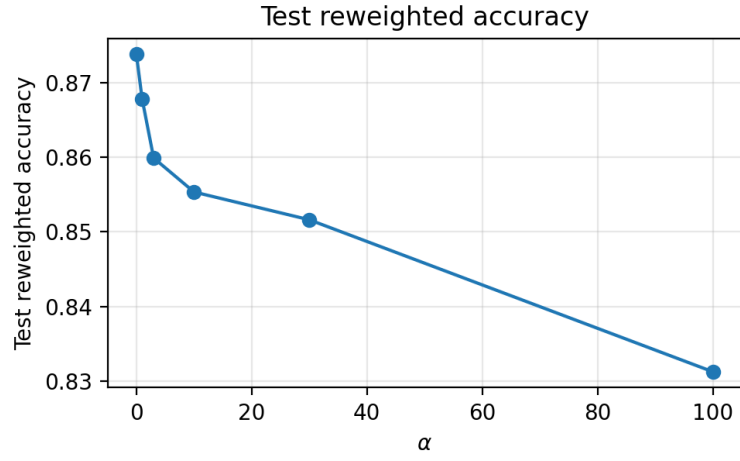
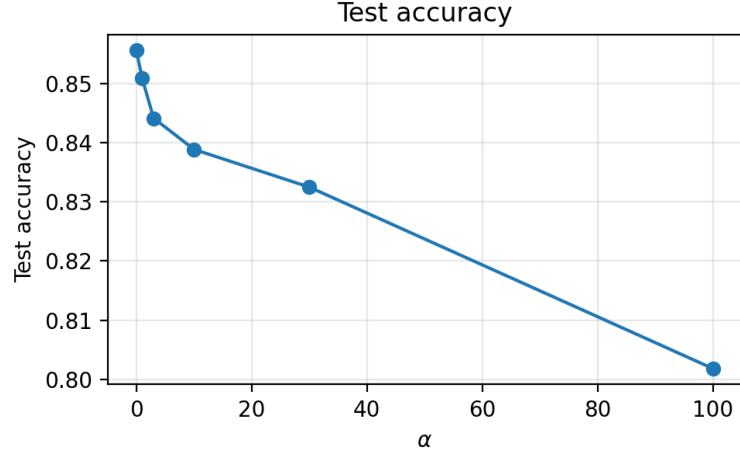
The minimized objective is binary cross-entropy plus αL_{DP} . Each score is a composition of smooth network operations and the sigmoid, both group means are linear in those scores, and $z \mapsto z^2$ is smooth. Thus L_{DP} has a gradient with respect to the network parameters. A thresholded prediction would be constant almost everywhere and would not provide this gradient. Batches missing either group use classification loss alone.

(b) Training-set sweeps showed that $\alpha = 0.1$ barely reduced the DP gap, while $\alpha \in \{300, 1000\}$ pushed both groups' positive rates to approximately zero. The useful comparison range, including the baseline, is therefore

$$\alpha \in \{0, 1, 3, 10, 30, 100\}.$$

All runs use seed 0 and the supplied training settings. The test curves are shown below; tabulated values are rounded to four decimals.

α	Accuracy	Reweighted accuracy	Δ_{DP}
0	0.8557	0.8738	0.1707
1	0.8509	0.8678	0.0958
3	0.8442	0.8599	0.0514
10	0.8389	0.8554	0.0211
30	0.8325	0.8516	0.0012
100	0.8018	0.8312	0.0040



Increasing α from 0 to 30 reduces test Δ_{DP} from 0.1707 to 0.0012 while accuracy falls by about 0.023 and reweighted accuracy by about 0.022. At $\alpha = 100$, the gap remains near zero, but accuracy falls further to 0.8018 because positive predictions become rare in both groups. The DP curve is not monotone: the gap is 0.0012 at $\alpha = 30$ and 0.0040 at $\alpha = 100$.

Question 2

Q2.1

Let X, R be independent Bernoulli($1/2$) variables and set $Y = X + R \pmod{2}$. The joint distribution has four atoms,

$$(X, R, Y) \in \{(0, 0, 0), (0, 1, 1), (1, 0, 1), (1, 1, 0)\},$$

each with probability $1/4$.

For every $x, r \in \{0, 1\}$, $P(X = x, R = r) = 1/4 = P(X = x)P(R = r)$, so $X \perp R$.

Now $P(Y = 0) = 1/2$, and the conditional outcomes are $(X, R) = (0, 0)$ and $(1, 1)$, each with probability $1/2$. Hence

$$P(X = 1, R = 1 \mid Y = 0) = \frac{1}{2},$$

whereas

$$P(X = 1 \mid Y = 0)P(R = 1 \mid Y = 0) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}.$$

The two quantities differ, so $X \not\perp R \mid Y$.

Q2.2

Let $P(A = 0) = P(A = 1) = 1/2$,

$$\begin{array}{lll} P(Y = 1 \mid A = 0) = \frac{1}{2}, & P(Y = 2 \mid A = 0) = \frac{1}{4}, & P(Y = 3 \mid A = 0) = \frac{1}{4}, \\ P(Y = 1 \mid A = 1) = \frac{1}{4}, & P(Y = 2 \mid A = 1) = \frac{1}{4}, & P(Y = 3 \mid A = 1) = \frac{1}{2}, \end{array}$$

and let $R = \mathbf{1}\{Y \neq 2\}$. The resulting joint atoms are

(A, Y, R)	$(0, 1, 1)$	$(0, 2, 0)$	$(0, 3, 1)$	$(1, 1, 1)$	$(1, 2, 0)$	$(1, 3, 1)$
Probability	$1/4$	$1/8$	$1/8$	$1/8$	$1/8$	$1/4$

Both groups and all three target values have positive probability. Since $P(Y = 1 \mid A = 0) = 1/2$ but $P(Y = 1) = 3/8$, it follows that $A \not\perp Y$.

Given either value of A , $P(R = 1 \mid A) = 3/4 = P(R = 1)$, so $R \perp A$. Given any value of Y , R is deterministic and therefore independent of A . Thus $R \perp A \mid Y$. Independence and separation hold simultaneously even though A and Y are dependent.

Q2.3

Let $p_a = P(g(X) = 1 \mid A = a)$ and $\delta = p_1 - p_0$, so $\Delta_{\text{DP}}(g) = |\delta|$. Define $h(X) = g(X)$ when $\delta \geq 0$, and $h(X) = 1 - g(X)$ when $\delta < 0$. In either case h is a function of X .

If $\delta \geq 0$, then

$$\begin{aligned} \text{RWAcc}_A(h) &= \frac{1}{2}(P(h = 0 \mid A = 0) + P(h = 1 \mid A = 1)) \\ &= \frac{1}{2}(1 - p_0 + p_1) = \frac{1}{2} + \frac{1}{2}\delta = \frac{1}{2} + \frac{1}{2}\Delta_{\text{DP}}(g). \end{aligned}$$

If $\delta < 0$, then

$$\begin{aligned}\text{RWAcc}_A(h) &= \frac{1}{2}(P(g = 1 \mid A = 0) + 1 - P(g = 1 \mid A = 1)) \\ &= \frac{1}{2} + \frac{1}{2}(p_0 - p_1) = \frac{1}{2} + \frac{1}{2}\Delta_{\text{DP}}(g).\end{aligned}$$

Therefore $\text{RWAcc}_A(h) \geq \frac{1}{2}\Delta_{\text{DP}}(g) + \frac{1}{2}$.

Assistance disclosure

Cursor, a generative-AI coding assistant, was used to implement the required functions and analysis script, run the supplied training and evaluation workflow, generate the figures, and draft this write-up. The code, metric checks, numerical results, and proofs were reviewed against the assignment definitions.